

RAJA KUMAR

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SUMMARY

Final-year B.Tech CSE student (CGPA 9.11) specializing in AI/ML, Generative AI, and intelligent systems. Experienced in building RAG applications, AI agents & vector databases, using Python, FastAPI, Docker, and CI/CD.

SKILLS

Programming Languages: Java, Python, C, SQL
Frameworks & Libraries: FastAPI, Flask, LangChain, LangGraph, Next.js, React, Express, TensorFlow
Tools & Platforms: AWS, Docker, Git, GitHub Actions, VS Code, PyCharm, Jupyter Notebook, CI/CD
AI/ML & Software Engineering: Machine Learning, Deep Learning, Generative AI, LLMs, RAG, System Design
Core Competencies: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

PROJECTS

- AI-Map: Local RAG & Vector Database Platform** | Python, HNSW, Ollama, Docker, GitHub Actions

GitHub

 - Built a Vector Database from scratch using HNSW ($O(\log N)$), KD-Tree, and Brute Force algorithms in Python.
 - Implemented an offline RAG pipeline with Ollama for document retrieval and AI-powered question answering.
 - Designed an interactive web interface with PCA visualization and benchmarking across multiple search algorithms and distance metrics.
 - Containerized the application with Docker Compose and automated testing and Docker builds using GitHub Actions CI/CD.
- Quant Agent: Multi-Agent Trading System** | Python, LangGraph, LangChain, Flask

GitHub

 - Developed a multi-agent AI trading platform using LangGraph for autonomous market analysis and trade decision orchestration.
 - Integrated GPT-4o, LLaVA, and TA-Lib indicators (RSI, MACD), reducing false trading signals by 30%.
 - Engineered scalable Flask REST APIs for real-time inference, agent communication, and consensus-based trade recommendations.
 - Automated containerized deployments using Docker, GitHub Actions, and CI/CD pipelines.
- MIRAI: Autonomous AI Combat System** | Python, FastAPI, FAISS, Ollama, Next.js

GitHub

 - Architected an AI backend inspired by NVIDIA ACE using FAISS vector retrieval, behavioral profiling, PyTorch, and XGBoost ensemble inference for adaptive boss intelligence.
 - Designed an online reinforcement learning-inspired feedback loop with telemetry-driven intent prediction, episodic memory, and adaptive strategy selection.
 - Built a FastAPI-based Cognitive OS using Ollama (Llama 3.2) for explainable AI reasoning and real-time tactical decision-making.

EDUCATION

KIIT UNIVERSITY

2023 – 2027

B.Tech in Computer Science and Engineering

CGPA: 9.11

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, Discrete Mathematics, Machine Learning, Data Mining

OTHER EXPERIENCE

SwasthyaPro

Jul 2025 – Sep 2025

Digital Marketing Intern (Remote)

- Improved organic search visibility and analyzed 1000+ monthly traffic insights using Google Analytics and SEMrush to optimize SEO performance.

ACHIEVEMENTS & CERTIFICATIONS

- Completed [AWS Cloud Architecting](#), [AICTE Data Science Internship \(Siemens\)](#), and [Data Science, AI/ML & GenAI](#).
- Solved 300+ DSA problems and secured 1st Runner-Up at KIIT Hackathon [UiPath Resume Screener](#).